

Chris' Very Excellent Acoustics Problems

Christopher Schaaf

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Good luck! :D

I. Drum Circles

1. Recall the wave speed equation for a (1D) string:

$$v = \sqrt{\frac{F}{\mu}}.$$

There is an analogous wave speed formula for a wave on a 2D surface, of the same form: $v = \sqrt{F'/\mu'}$. However, F' and μ' have different dimensions. What are they?

2. The same as (1), but for a 3D wave (say, a sound wave.) What important physical measurement has the dimensions of F' ?

3. Using the results in (2), and a formula from Chemistry, prove the *temperature equation*:

$$v = v_0 \sqrt{\frac{T}{T_0}}.$$

4. Recall that a standing wave on a string of length L must have wavelengths of the length $\frac{2L}{n}$, with n a natural number. For any such standing wave, give an equation $a(x, t)$ that gives the amplitude of any given point on such a string. Express your answer in terms of n , L , the frequency f , and the amplitude A .

5. Now we will derive an analogous formula for a standing wave in 2D. Imagine a square drum of side length r_0 . Each point in this drum is free to oscillate up and down. (Please watch this video for a visual.) The drum is vibrated at an arbitrary frequency. Determine formulas for the possible standing waves on this surface.

Hint: a standing wave can be written with independent spatial and temporal components, i.e. $f(x, y, t) = g(x, y) \cdot h(t)$. For a standing wave on (say) a square drum, the points boundary of the drum must have 0 amplitude, i.e. $g(x, y) = 0$. You might also need the 2D wave equation

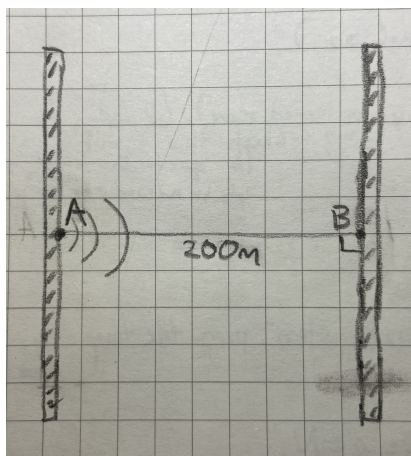
$$\frac{\partial^2}{\partial x^2} \psi + \frac{\partial^2}{\partial y^2} \psi = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \psi.$$

6. Replace the square drum with a circular drum of radius r_0 . Determine the first non-trivial standing wave on such a drum, i.e. one with a trivial angular component.

II. Miroirs

A *Lindo reflector* is a material with the following curious property: any sound waves that interact with it reflect off with exactly half their original amplitude (the reflector absorbs the rest.)

Dr. Meadowcroft has constructed the following experimental setup:



Here, two flat and parallel walls, made of Lindo reflector, are placed 200m away from each other. (In the diagram, the walls appear as lines; in reality it is better to treat them as flat planes which extend “into the page.”) Points A and B are chosen such that line AB is perpendicular to each of the walls. Meadowcroft attaches a speaker to point A. The speaker continuously emits a sound of wavelength $\lambda = 1\text{m}$, wave speed $v = 400\text{m/s}$, and amplitude $A = 1\text{W/m}^2$ evenly in all directions. The sound reflects off of the walls and interferes with itself. Meadowcroft measures the sound intensities of each point on both walls.

1. What is the sound intensity on point A after 30 seconds?
2. What is the sound intensity on point B after 60 seconds?
3. Determine the distance between point B and the first destructive interference on the right wall, after 60 seconds. (Note that the destructive interference forms a ring.)

Meadowcroft repeats this experiment. However, instead of recording the sound intensity after 60 seconds, he measures the total energy absorbed by the walls at each given point. (For simplicity, assume that there is no heat transfer between different points on the walls. The only absorption is through the sound.)

4. Draw a circle of radius 0.001m on the right wall, around point B. What is the total energy absorbed by the wall, inside this circle, after 60 seconds?
5. Same as (4), with the circle drawn around a point of destructive interference found in (3).

N.B.: when sound propagates in all directions, its intensity decreases as the distance squared.